Pritam

Portfolio: prtmxio

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Technical Skills

Programming Languages: C/C++, Python, Java, Verilog
Embedded & Robotics: ESP-IDF, ROS 2, Arduino, ForgeFPGA, MQTT, PID Control, Firmware Development
AI / ML: PyTorch, HuggingFace Transformers & Diffusers, CLIP, OpenCV, Scikit-learn, CNN, Vector Search (HNSW)
Tools: Git, Linux, KiCad, VS Code, (neo)Vim, Gazebo, URDF, Onshape API

Experience

Embedded Systems Intern **Mar 2026 – Present**
Glazia *Faridabad, Remote*

- Built **Glazia Home Secure**, a dual-ESP32-S3 IoT security system with fingerprint auth, LVGL touch UI, and a dedicated camera coprocessor for capture.
- Implemented real-time **WebRTC** video streaming (H.264, DTLS-SRTP) over STUN/TURN, sustaining **15 fps** at VGA end-to-end.
- Built an encrypted **ESP-NOW** sensor mesh (AES-128-GCM) supporting **10 concurrent nodes** with nonce-based replay protection.
- Designed dual-factor device authentication (MAC + server secret), enforced across all hub-server routes.

Embedded Application Developer (Intern) **Jul 2025 – Dec 2025**
PCB CUPID *Bangalore, Remote*

- Co-built **Circuit Spell** (spell.pcbcupid.com), an AI agent generating full PCB designs from natural language using **50+** proprietary Glyph modules.
- Prototyped EDA subsystems for constraint-based component placement and auto-routing.
- Developed multi-sensor IoT firmware in C/C++ on Glyph boards for industrial control.

Projects

meshflow | *Python, ROS 2, Gazebo, Onshape API, URDF/Xacro* | **June 2026**

- Built a CLI tool converting Onshape CAD assemblies into simulation-ready ROS 2 packages (URDF, xacro, Gazebo plugins) via the Onshape REST API.
- Designed **KinematicDAG**, a geometry-only classifier inferring robot type and sensor kind from URDF joint structure, needing zero manual naming conventions.
- Auto-generated Gazebo/`ros2_control` plugins with wheel geometry measured via **trimesh**, verifying lidar throughput at **20Hz** in simulation.

synapse | *Transformers, Diffusers, HNSW* | **Mar 2026**

- Built a local multimodal pipeline (SmolVLM → CLIP → SD-Turbo) for image-conditioned generation, fitting entirely within a **4 GB VRAM** budget.
- Implemented **HNSW** ANN indexing for CLIP-based retrieval, achieving **1ms** query latency regardless of library size.
- Achieved end-to-end latency of **6–10s** (prompt) and **12–14s** (image) on an RTX 2050 via sequential model load/unload.

Education

Deenbandhu Chhotu Ram University of Science and Technology **2023 – Present**
Bachelor of Technology in ECE; CGPA: 8.49/10 *Sonipat, Haryana*

Achievements

IEEE YESIST12 2026 : International Grand Finale Qualifier | *Team Member & Software Developer* **2024**

- Built ARGUS, an autonomous AIOps platform combining ML-based anomaly detection (Isolation Forest + z-score) with LLM-driven root cause analysis to self-heal cloud microservice failures in real time.
- Abstract selected among 2,000+ global submissions for the international grand finale in Jakarta, Indonesia.